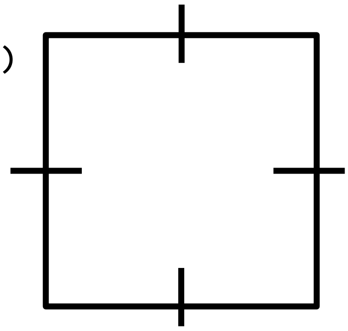
Using inequality notation for errors in calculations

Task A: Error intervals (addition and multiplication)

1) Work out the error interval for the perimeter of these shapes:

a)



$2.5 \le l < 3.5$

$2.5 \times 4 \le p < 3.5 \times 4$

$10 \le p < 14$

b)

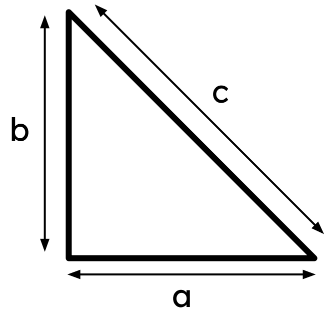
$2.25 \le w < 2.35$

$7.85 \le l < 7.95$

$2(2.25+7.85) \le p < 2(2.35+7.95)$

$20.2 \le p < 20.6$

c)



$4.5 \le a < 5.5$

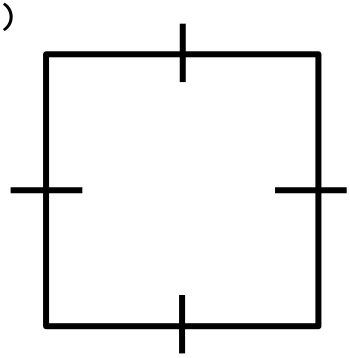
$11.5 \le b < 12.5$

$12.5 \le c < 13.5$

$28.5 \le p < 31.5$

2) Work out the error interval for the area of these shapes:

a)

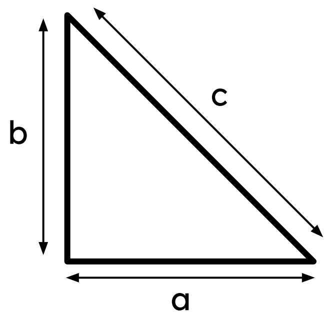


$4.25 \le l < 4.35$

$4.25^2 \le a < 4.35^2$

$18.0625 \le a < 18.9225$

b)



$4.5 \le a < 5.5$

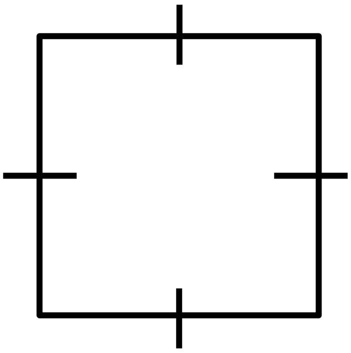
$11.5 \le b < 12.5$

Lower area : $0.5 \times 4.5 \times 11.5$

Upper area : $0.5 \times 5.5 \times 12.5$

$25.875 \le area < 34.375$

3) Given the error interval of the **area**, can you work out the rounded lengths and the degree of accuracy?



Square with lengths
8.3 cm (1 d.p)

$\sqrt{68.0625} \leq l < \sqrt{69.7225}$

$8.25 \leq l < 8.35$

$\boxed{68.0625 \text{ cm}^2 \leq area < 69.7225 \text{ cm}^2}$

Task B: Error intervals in calculations

1) A rope was originally 5.3 cm (1 d.p). A piece of 3.8 cm (1 d.p) was cut from the rope. What is the error interval for the remaining length?

Original length
$5.25 \leq l < 5.35$

Cut length
$3.75 \leq c < 3.85$

Upper limit remaining : 5.35 - 3.75 = 1.6

Lower limit remaining : 5.25 - 3.85 = 1.4

Remaining length
$1.4 < remaining < 1.6$

2) A coffee had 330 ml (the nearest integer) and Lucas drank 32 ml (2 s.f.). What is the error interval for the remaining coffee?

Original full coffee
$329.5 \leq c < 330.5$

Amount drank
$31.5 \leq d < 32.5$

Upper limit remaining : 330.5 - 31.5 = 299

Lower limit remaining : 329.5 - 32.5 = 297

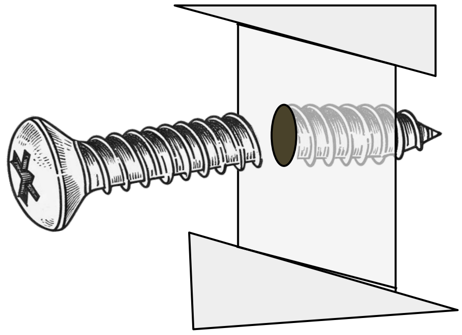
Remaining coffee
$297 < coffee < 299$

3) A screw has broken off from a wall.

Originally the screw was 4.6 mm (1 d.p).

The broken off piece measures 3.4 mm (1 d.p).

Work out the error interval length of the screw which is lodged in the wall.



Original full screw
$4.55 \leq s < 4.65$

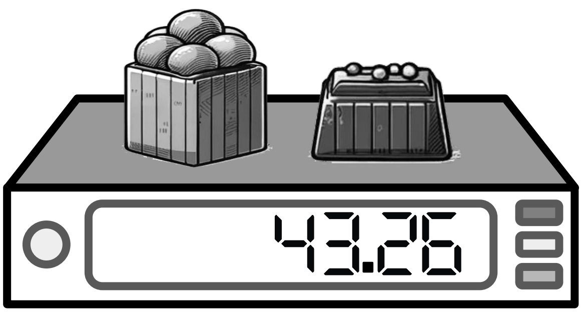
Broken off
$3.35 \leq b < 3.45$

Upper limit lodged in wall : 4.65 - 3.35 = 1.3

Lower limit lodged in wall : 4.55 - 3.45 = 1.1

Lodged in wall

$1.1 < l < 1.3$



4) A box of sweets and a chocolate weigh 43.26 g correct to 2 d.p.

If the chocolate alone weighs 21.14 g (2 d.p), what is the error interval for the mass of the box of sweets?

Total mass
$43.255 \leq m < 43.265$

Chocolate
$21.135 \leq c < 21.145$

Upper limit sweets : 43.265 - 21.135 = 22.13

Lower limit sweets : 43.255 - 21.145 = 22.11

Mass of Sweets in box

$22.13 < s < 22.11$